

Dual-Material Double-Gate SOI n-MOSFET: Gate Misalignment Analysis

Rupendra Kumar Sharma, Ritesh Gupta, Mridula Gupta, *Member, IEEE*, and R. S. Gupta, *Senior Member, IEEE*

Abstract—The dual-material double-gate (DMDG) silicon-on-insulator (SOI) metal-oxide-semiconductor field-effect transistor (MOSFET) is the leading contender for sub-100-nm devices because it utilizes the benefits of both double-gate and dual-material-gate structures. One major issue of concern in the DMDG-MOSFET is the alignment between the top and the bottom gate that critically influences the device performance. In this paper, we have investigated the effects of gate misalignment in the DMDG SOI n-MOSFET. In this regard, analytical modeling and extensive simulations have been carried out to analyze the gate misalignment effects on device performance like surface potential, electric field, threshold voltage, subthreshold slope, drain-induced barrier lowering, drain current, and transconductance. Considering the fact that gate misalignment can occur on any side of the gate, both source- and drain-side misalignments have been discussed. Analytical and simulated results are found to be in good agreement, which authenticate our proposed model for the DMDG structure.

Index Terms—Drain current, dual-material double gate (DMDG), gate misalignment, subthreshold slope, threshold voltage, transconductance, 3-D ATLAS device simulation.

I. INTRODUCTION

SINCE THE last three decades, continuous scaling of metal-oxide-semiconductor field-effect transistor (MOSFET) devices has been successfully propelled by the rapid advancement of device engineering and fabrication technologies for gaining better performance and higher packing density [1], [2]. As CMOS scaling approaches the limit imposed by gate oxide tunneling, the double-gate (DG) MOSFET is becoming the subject of intense VLSI research, owing to its ability to be scaled to the shortest channel length possible for a given gate oxide thickness [3]–[6]. Excellent short-channel-effect (SCE) immunity, improved threshold-voltage roll-off, higher drain current, higher transconductance, and an ideal subthreshold slope have been reported by various studies on this device [7]–[16].

As MOSFET dimensions are scaled down to the sub-100-nm regime, threshold-voltage reduction with decreasing channel lengths and drain-induced barrier lowering (DIBL) become important issues that should be addressed while providing immunity against SCEs. Similar issues have also been observed in sub-100-nm-gate-length DG devices. Gate work-function

engineering, i.e., dual-material-gate (DMG) MOSFET, was proposed to enhance immunity against SCEs [17]–[20]. This DMG MOSFET has two metals M_1 and M_2 of different work functions $q\phi_{M1}$ and $q\phi_{M2}$, respectively. These two metals are amalgamated together laterally and provide simultaneous increases in transconductance and suppressed SCEs due to a step in the surface-potential profile in comparison to a single-gate MOSFET. A similar DMG structure was also proposed by Reddy and Kumar [21], [22] to suppress the SCEs in DG devices. This dual-material double-gate (DMDG) structure reduces the peak electric field near the drain end, increases the drain breakdown voltage, improves the transconductance, and reduces the drain conductance and desired threshold-voltage roll-up for channel lengths far below 100 nm. Chiang and Chen [23] also reported an analytical model for the DMDG structure and discussed several benefits such as improved SCEs, DIBL effects, etc. In spite of the several benefits, the major issue that remains in the DG MOSFET is its susceptibility to process variations that affect the eventual circuit performance. Among them, the most significant concern that gives rise to variations in the DG device performance is the misalignment between the top and the bottom gate.

The misalignment between the gates would affect several device properties of DG MOSFETs, specifically SCEs, subthreshold slope, DIBL, etc. The bottom gate can either shift to the source side (DGD) or to the drain side (DGS), so a certain part of the channel region toward either the source or drain end is controlled by only one gate. Nonideal effects, introduced by this gate misalignment, can emerge from either the nonoverlap region or the overlap region. The bottom channel of the device under the nonoverlap gate region is weakly controlled by the bottom gate through the fringing electric field. This weakly controlled region of the bottom channel introduces large series resistance due to the less inversion carriers and hence affects device operation. On the other hand, the overlap region introduces additional gate leakage current through direct tunneling [24]. The subthreshold characteristics become worse as the misalignment increases due to a degradation of the gate electrostatic control, as well as a reduced total gate capacitance in comparison to the drain capacitance. This decreased electrostatic control results in increased I_{off} , thereby showing diminished channel controllability by the gates. In terms of SCEs, aligned transistors demonstrate the best control, while highly misaligned MOSFETs operate like single gates. Extensive studies have already been reported by many authors [25]–[27] regarding gate misalignment in the DG structure and its effects on device performance. Misalignment causes threshold-voltage increase and drain-current and transconductance

Manuscript received January 14, 2009; revised March 17, 2009. Current version published May 20, 2009. This work was supported by the Defence Research and Development Organisation (DRDO), Ministry of Defence, Government of India. The review of this paper was arranged by Editor M. J. Kumar. The authors are with the Semiconductor Devices Research Laboratory, Department of Electronic Science, University of Delhi South Campus, New Delhi 110021, India (e-mail: rsgu@bol.net.in; rupendra1984@gmail.com).

Digital Object Identifier 10.1109/TED.2009.2019695

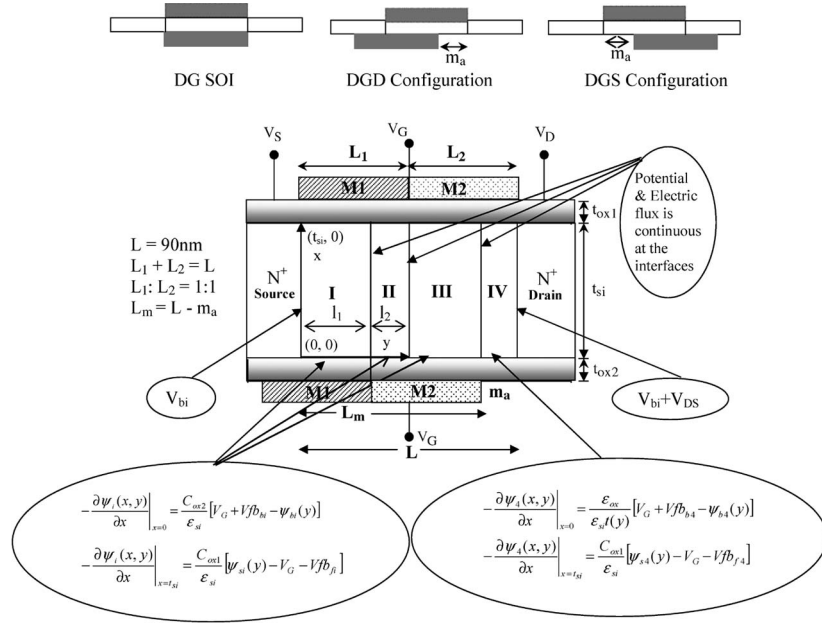


Fig. 1. Schematic structure of the DMDG SOI MOSFET having channel length $L = 90$ nm, $t_{si} = 10$ nm, $t_{ox1} = t_{ox2} = 3$ nm, $N_A = 3 \times 10^{18}$ cm $^{-3}$, $\phi_{M1} = 4.8$ eV, and $\phi_{M2} = 4.5$ eV with boundary conditions and gate misalignment of m_a .

decreases. We have also reported the graded channel architecture as a solution for misaligned DG structures [28].

Due to the 2-D and 3-D nature of physical effects governing the electrical behavior of VLSI MOS transistors, it is very difficult to obtain a closed-form analytical formulation that is valid in all operating regions of interest. However, one can still obtain closed-form analytical models, based on device physics that are generally valid only over a limited region of device operation. In this paper, an analytical model has been developed to show the effect of gate misalignment on the surface potential of the DMDG structure. In this model, equations derived directly from device physics are used to represent the subthreshold region of device operation. The model represents a first-order device behavior fairly accurately and describes the relationship between the physical process and geometric structure, as well as the electrical device behavior. Intensive simulation using a 3-D ATLAS device simulator [29] has been done to verify the analytical results in the subthreshold region of device operation and to also provide the device characteristics under ON conditions.

II. THEORETICAL CONSIDERATIONS

The basic device structure of the misaligned DMDG SOI n-MOSFET used in this paper is shown in Fig. 1 for both drain- and source-side gate misalignments. It consists of a silicon film of thickness $t_{si} = 10$ nm, having a p-type channel impurity concentration N_A . This silicon film is sandwiched between the gate insulator [SiO $_2$ ($\epsilon_{ox} = 3.9$ and $t_{ox} = 3$ nm)] of equal thickness and the amalgamated gate structure assembly of different metal work functions ($\phi_{M1} = 4.8$ eV, $\phi_{M2} = 4.5$ eV, $L = 90$ nm, and $L_1 : L_2 = 1 : 1$) on both sides. The N $^+$ ($= 10^{20}$ cm $^{-3}$) source/drain regions are considered to be of sufficient length, so that gate misalignment can vary up to 50% of the gate length. For the DMDG structure, the metal work function of the control

gate is assumed to be of M_1 ($= 4.8$ eV), and that of the screening gate is considered to be of M_2 ($= 4.5$ eV). The analysis has been performed for the misaligned DMDG architecture by splitting the channel into four different regions, i.e., three for different combinations of metal gate work functions and one for the ungated region. Potentials and charges are calculated solving Poisson's equation using a classical density of states. Quantum-mechanical effects are significant for thicknesses that are smaller than 5 nm and for channel lengths that are below 10 nm [30]. In this paper, we have taken channel length $L = 90$ –30 nm and $t_{si} = 10$ nm and focused our attention on the gate misalignment effect on device performance and neglect the quantum-mechanical effects.

III. FRONT AND BACK SURFACE POTENTIALS

The 2-D potential distribution is obtained by solving the following 2-D Poisson's equation for all the four regions with appropriate boundary conditions, as shown in Fig. 1:

$$\frac{\partial^2 \psi(x, y)}{\partial x^2} + \frac{\partial^2 \psi(x, y)}{\partial y^2} = \frac{qN_A}{\epsilon_{si}} \quad (1)$$

where ϵ_{si} is the silicon permittivity, q is the electronic charge, and $\psi(x, y)$ is the 2-D potential distribution in the silicon film. In the DMDG structure, we have two materials with different work functions, which are amalgamated together laterally. Therefore, the flatband voltage for two gates would be different, as it depends on the metal gate work functions of metals M_1 and M_2 , respectively. The back- and front-gate flatband voltages Vfb_{bi} and Vfb_{fi} are given by

$$Vfb_{bi} = \phi_{Mbi} - \phi_S \quad Vfb_{fi} = \phi_{Mfi} - \phi_S \quad (2)$$

where i varies from one to three for regions I, II, and III, respectively. The semiconductor work function can be written

as $\phi_S = \chi_S + (E_g/2) + \phi_f$, where $\phi_f = (kT/q) \ln(N_A/n_i)$ is the bulk potential, E_g is the silicon band gap, and χ_S is the electron affinity. Using the boundary conditions shown in Fig. 1, the analysis for regions I, II, and III are the same, except for the different combinations of metal gate work functions. The back surface potential (ψ_{bi}) and the front surface potential (ψ_{si}) for the three regions are obtained by using the similar approach proposed by Sun and Kuo [26] and are given by

$$\psi_{bi}(y) = U_{1i} \exp(\sqrt{A_{0i}}y) + U_{2i} \exp(-\sqrt{A_{0i}}y) - \frac{A_{1i}}{A_{0i}} \quad (3)$$

$$\psi_{si}(y) = U_{3i} \exp(\sqrt{A_{0i}}y) + U_{4i} \exp(-\sqrt{A_{0i}}y) - D_i \quad (4)$$

where A_{0i} , A_{1i} , and D_i are defined in (5a), (5b), and (5c), respectively, shown at the bottom of the page.

Similarly, for region IV, the front surface potential (ψ_{s4}) and the back surface potential (ψ_{b4}) are obtained and given as

$$\psi_{b4}(y) = g_1 \exp\left[\frac{\gamma}{k}(y - L_m)\right] + g_2 \exp\left[-\frac{\gamma}{k}(y - L_m)\right] + \frac{B_1}{B_0} \quad (6)$$

$$\psi_{s4}(y) = g_3 \exp\left[\frac{\gamma}{k}(y - L_m)\right] + g_4 \exp\left[-\frac{\gamma}{k}(y - L_m)\right] + D_4 \quad (7)$$

where γ , B_0 , B_1 , k , and D_4 are given in (8a) and (8b), shown at the bottom of the page.

All coefficients U_{1i} , U_{2i} , U_{3i} , U_{4i} , g_1 , g_2 , g_3 , and g_4 can easily be evaluated using the boundary conditions shown in Fig. 1 for both DGD and DGS structures. σ_2 is an empirical correction factor to account for the nonsymmetrical structure in the ungated region [26]. κ_S is a parameter that relates the

derivative of lateral electric field at any depth in the silicon film to the derivative of lateral electric field at the front Si-SiO₂ interface [31].

IV. RESULTS AND DISCUSSION

The results for surface potential have been calculated through an analytical model, whereas all the device characteristics have been obtained by using a 3-D ATLAS device simulator. In the TCAD simulation, the drift-diffusion model has been used, which takes into account the attractive feature that it does not introduce any independent variables besides electrostatic potential and electron and hole concentrations. We adopt the CONMOB model for low field mobility. In that model, the mobility is only related to the doping density. In the high field, we adopt the FLDMOB model for velocity saturation, which takes into account the dependence on parallel electric field in the direction of the current flow. For the same reason, we chose Gummel's method (or the decoupled method), along with Newton's method (or the fully coupled method), to solve those equations included in the drift-diffusion model. The threshold voltage is obtained from $I_{DS} - V_{GS}$ characteristics and is considered to be that value of gate voltage for which the drain current approaches 10^{-7} A/ μ m. DIBL has been calculated as differences of threshold voltage at $V_{DS} = 50$ mV and 1.0 V. Subthreshold slope is defined as the change in gate voltage V_{GS} required to reduce subthreshold current I_{DS} by one decade and extracted at $V_{DS} = 50$ mV.

We compare in Fig. 2(a)–(d) the simulated and analytical front and back surface potential distributions along the normalized channel position for the SMDG and DMDG structures with different gate misalignments. The analytical results are in good agreement with the simulated results, and from these

$$A_{0i} = \frac{2\kappa_{si}}{t_{ox2}} \left(t_{ox2} + \frac{\varepsilon_{ox}}{\varepsilon_{si}} t_{si} + t_{ox1} \right) \frac{t_{si}^2 \left(1 + \frac{2\varepsilon_{si} t_{ox1}}{\varepsilon_{ox} t_{si}} \right)}{\quad} \quad (5a)$$

$$A_{1i} = \kappa_S \left[\frac{qN_A}{\varepsilon_{si}} - \frac{2}{t_{ox2}} \left(\left(t_{ox2} + \frac{\varepsilon_{ox}}{\varepsilon_{si}} t_{si} + t_{ox1} \right) V_G + \left(t_{ox1} + \frac{\varepsilon_{ox}}{\varepsilon_{si}} t_{si} \right) V_{fb_{bi}} \right) + 2V_{fb_{fi}} \right] \frac{t_{si}^2 \left(1 + \frac{2\varepsilon_{si} t_{ox1}}{\varepsilon_{ox} t_{si}} \right)}{\quad} \quad (5b)$$

$$D_i = \frac{\frac{1}{t_{ox2}} \left\{ \sigma_1 \left(1 + \frac{2\varepsilon_{si}}{\varepsilon_{ox} t_{si}} \right) \psi_{bi}(y) + \left(1 - \frac{t_{ox2}}{t_{ox1}} \right) V_G + V_{fb_{bi}} \right\} - \frac{V_{fb_{fi}}}{t_{ox1}}}{\left[\frac{1}{t_{ox1}} + \frac{2\varepsilon_{si}}{\varepsilon_{ox} t_{si}} \right]} \quad (5c)$$

$$\gamma = \sqrt{-B_0} \quad B_0 = 2\kappa_S k^2 \alpha_0 \quad B_1 = \kappa_S k^2 \left[\frac{qN_A}{\varepsilon_{si}} - 2\beta_0 \right] \quad k = \frac{m_a}{\sinh \left[\cosh^{-1} \left(\frac{t_{ox2} + t_{g2}}{t_{ox2}} \right) \right]} \quad (8a)$$

$$D_4 = \frac{\frac{1}{t(y)} \left\{ \sigma_2 \left(1 + \frac{2\varepsilon_{si} t(y)}{\varepsilon_{ox} t_{si}} \right) \frac{B_1}{B_0} + \left(1 - \frac{t(y)}{t_{ox1}} \right) V_G - V_{fb_{b4}} \right\} + \frac{V_{fb_{f4}}}{t_{ox1}}}{\left[\frac{1}{t_{ox1}} + \frac{2\varepsilon_{si}}{\varepsilon_{ox} t_{si}} \right]} \quad (8b)$$

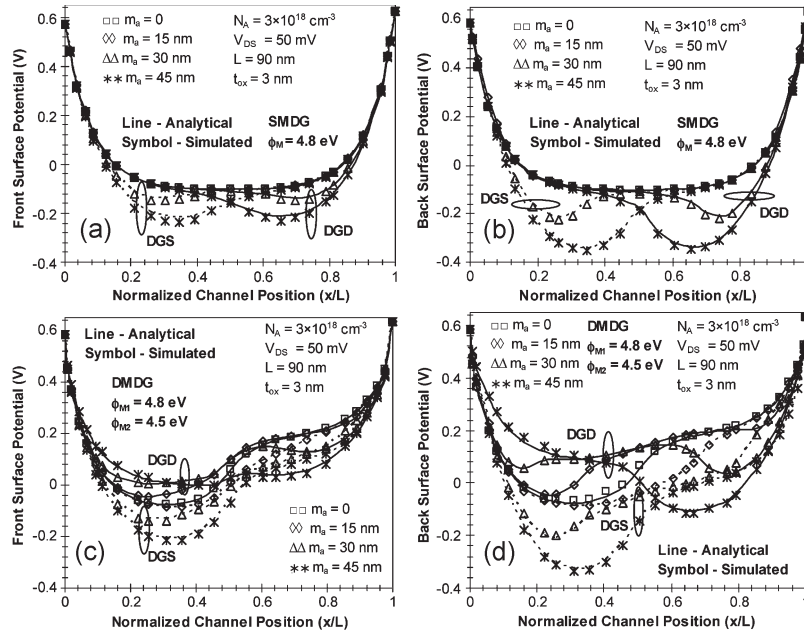


Fig. 2. (a)–(d) Variation of surface potential with lateral direction for SMDG and DMDG structures at $V_{DS} = 50 \text{ mV}$ and $V_{GS} = 0.2 \text{ V}$. (a) Front surface potential of SMDG. (b) Back surface potential of SMDG. (c) Front surface potential of DMDG. (d) Back surface potential of DMDG.

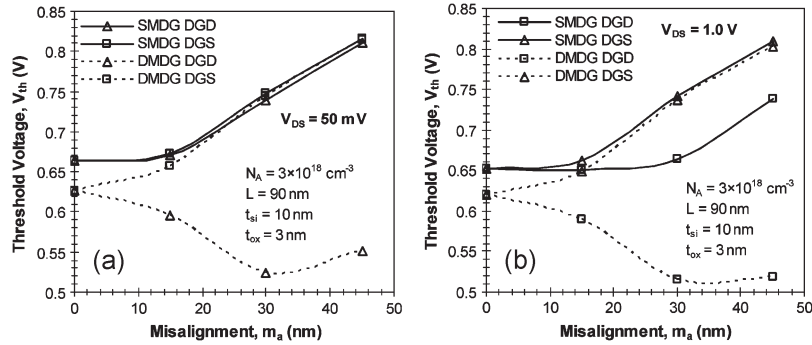


Fig. 3. Threshold-voltage variation with gate misalignment of SMDG and DMDG SOI MOSFETs for $L = 90 \text{ nm}$. (a) $V_{DS} = 50 \text{ mV}$. (b) $V_{DS} = 1.0 \text{ V}$.

curves, one can see that the model effectively models the surface potential of the SMDG and DMDG structures in the gate overlap, as well as in the ungated region.

The threshold-voltage variation with gate misalignment for both the SMDG and DMDG structures is shown in Fig. 3(a) for $V_{DS} = 50 \text{ mV}$. In the SMDG structure, for $m_a = 0$, the device threshold voltage is 0.664 V . With the increase in misalignment by 15 nm ($L/6$), 30 nm ($L/3$), and 45 nm ($L/2$) toward the drain side (DGD), the threshold voltage increases by 0.95% , 11.42% , and 22.12% , respectively. For the DGS structure, these values increase by 1.37% , 12.56% , and 22.80% , respectively. The cause of this threshold-voltage variation has been explained through the variation of front and back surface potentials along the channel, as shown in Fig. 2(a) and (b). Threshold voltage is usually controlled by the minima of the channel potential. When misalignment occurs, a certain part of the channel region is controlled by only one gate, i.e., an increase in effective silicon thickness decreases the minimum surface potential, thus increasing the threshold voltage of SMDG devices. In the DMDG structure, for $m_a = 0$, the threshold voltage is found to be 0.6266 V . With the increase in misalignment by 15 nm ($L/6$), the threshold voltage changes by 5.06% . This value

changes by 16.28% with the increase in gate misalignment by 30 nm ($L/3$), and it is further changed by 11.99% with the increase in misalignment by 45 nm ($L/2$). The threshold voltage continues to decrease with the increase in misalignment up to 30 nm . However, after 30-nm gate misalignment, the threshold voltage subsequently increases with the increase in misalignment. This is because the threshold voltage of the top channel is controlled by the front surface potential, whereas the threshold voltage of the bottom channel is controlled by the back surface potential. In DG devices, the top and bottom channels are in parallel. Therefore, the overall threshold voltage of the DG device is represented by the lowest threshold voltage out of the top and bottom channels. When misalignment occurs, an ungated region in the bottom channel leads to an increase in threshold voltage of the device. Thus, the overall threshold voltage of the device is controlled only by the front surface potential, i.e., top channel of the device. The variation in threshold voltage (front surface potential) can be explained through Fig. 1. In this figure, at $m_a = 0$, there exist only two regions under the different metal work-function assembly and threshold voltage, i.e., the surface potential profile of both channels is the same, and it is controlled by the higher work-function gate,

i.e., counter gate region. When misalignment occurs, the overall device performance can be explained by dividing the channel into four different regions. These regions can be treated as four different devices attached in series, and the overall threshold voltage of the device is represented by the device having the maximum threshold voltage. The region I of the DG device has a higher work function on both sides of the channel, region II has a higher/lower work function at the top/bottom channel, region III has a lower work function on both sides of the channel, and region IV is an ungated device at the bottom channel having a lower gate work function at the top channel. Now, the threshold voltage of the higher work-function region in the DG device is greater than that of the region having a lower work function. According to Wong *et al.* [32], if we choose the combination of higher–lower gate work function at the top–bottom channel, the threshold voltage of the device decreases in comparison to a device having a higher–higher gate work function and increases in comparison to a device having a lower–lower work function. Furthermore, the ungated region always has a higher threshold voltage in comparison to the gated region because of the increase in effective silicon thickness. It can be argued that $V_{th1} > V_{th2} > V_{th3}$, and V_{th4} can be either greater or lesser than V_{th1} , depending on the length of the ungated region. Moreover, misalignment causes a decrease in the minimum back surface potential in the ungated region, thereby decreasing the minimum front surface potential of that region. Lowering of surface potential in the ungated region affects the minimum surface potential in other regions, until there is a barrier formed by higher–lower gate work function. For $m_a = 15$ nm in the DGD configuration, the threshold voltage of the device decreases due to the emergence of region II and the reduction in the length of region I. The increase in misalignment by $m_a = 30$ nm causes a further decrease in threshold voltage owing to the increase in the length of region II. A further increase in misalignment by $m_a = 45$ nm shows an ambiguous behavior, i.e., having a higher threshold voltage in comparison to $m_a = 30$ nm. This is due to the fact that the back surface potential in the ungated region eliminates the potential barrier in the front channel and transmits the effect of the ungated region into the other regions, resulting in lowering of surface potential in region II. Also, region I is completely eliminated, and the threshold voltage of the device is controlled by region II. Fig. 3(b) shows the threshold-voltage variation with gate misalignment for both the SMDG and DMDG structures at $V_{DS} = 1.0$ V. In the SMDG structure, the effect of misalignment decreases in the DGD configuration at higher drain voltage, whereas in the DGS configuration, the effect of misalignment remains unaffected. This is because of the increase in the fringing field effect that arises at the drain side for the DGD configuration, in the presence of higher drain voltage.

DIBL and subthreshold swing are two important parameters to determine the short-channel immunity of a device, as shown in Figs. 4 and 5, respectively. The DGD configuration of the SMDG structure exhibits very high DIBL effects in comparison to the DGS configuration. This is due to the high drain bias for the DGD configuration in comparison to the DGS one. The DMDG structure shows lesser DIBL effect in comparison to the SMDG structure for both DGD and DGS configurations

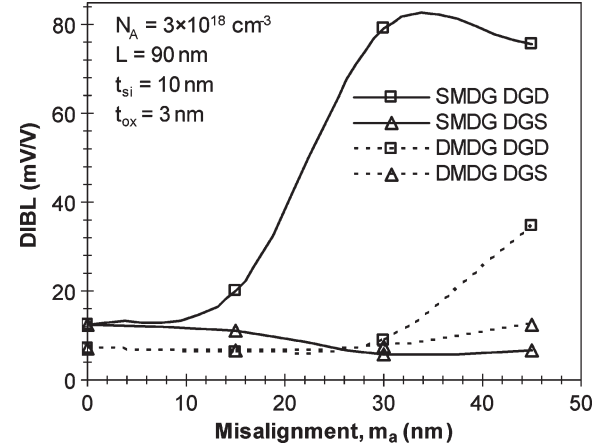


Fig. 4. DIBL variation with gate misalignment of SMDG ($\phi_M = 4.8$ eV) and DMDG ($\phi_{M1} = 4.8$ eV and $\phi_{M2} = 4.5$ eV) SOI MOSFETs for $V_{DS} = 50$ mV.

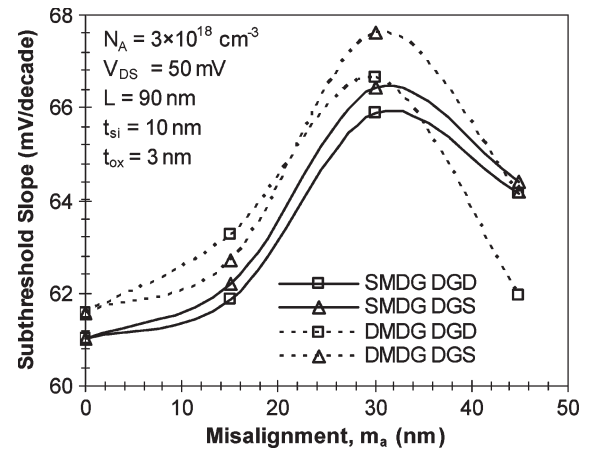


Fig. 5. Subthreshold-slope variation with gate misalignment of SMDG ($\phi_M = 4.8$ eV) and DMDG ($\phi_{M1} = 4.8$ eV and $\phi_{M2} = 4.5$ eV) SOI MOSFETs for $V_{DS} = 50$ mV.

up to $m_a = 15$ nm. The DGD configuration of DMDG also shows a higher DIBL for larger gate misalignment in comparison to the DGS configuration. However, both DGD and DGS configurations of the DMDG structure exhibits higher values of subthreshold slope in comparison to the SMDG structure, as shown in Fig. 5.

The subthreshold drain current characteristics of SMDG and DMDG MOSFETs are shown in Fig. 6(a) and (b), respectively. With the increase in misalignment, the DGD and DGS configurations of the SMDG structure demonstrate an almost similar degradation in subthreshold drain-current characteristics. The DGS configuration of the DMDG structure shows a slightly large variation in subthreshold drain current in comparison to the SMDG structure. However, the DGD configuration of the DMDG structure shows conflicting behaviors, i.e., with the increase in misalignment, the subthreshold drain current increases.

Fig. 7(a) and (b) shows the simulated variations of drain current and transconductance with gate misalignment for the SMDG and DMDG structures under ON conditions. As misalignment increases, drain current and transconductance decrease for the SMDG and DMDG structures. This is because,

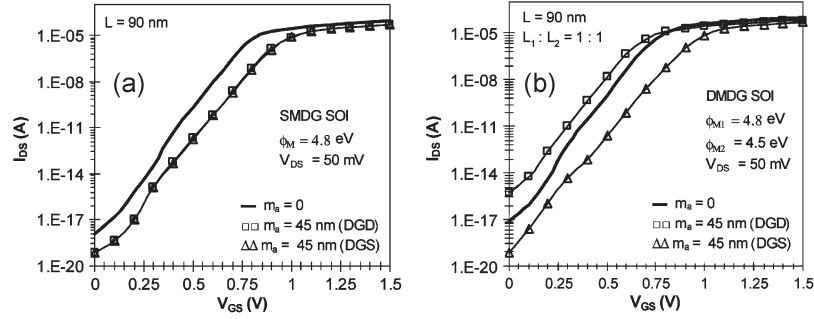


Fig. 6. Subthreshold drain-current variation with gate voltage for SMDG and DMDG SOI MOSFETs at $V_{DS} = 50$ mV. (a) SMDG SOI MOSFET. (b) DMDG SOI MOSFET.

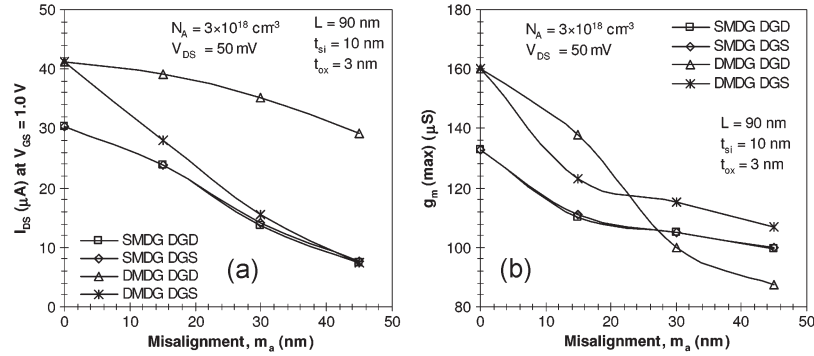


Fig. 7. I_{DS} and g_m (max) variation with gate misalignment of SMDG ($\phi_M = 4.8$ eV) and DMDG ($\phi_{M1} = 4.8$ eV and $\phi_{M2} = 4.5$ eV) SOI MOSFETs. (a) $V_{DS} = 50$ mV and $V_{GS} = 1.0$ V. (b) $V_{DS} = 50$ mV.

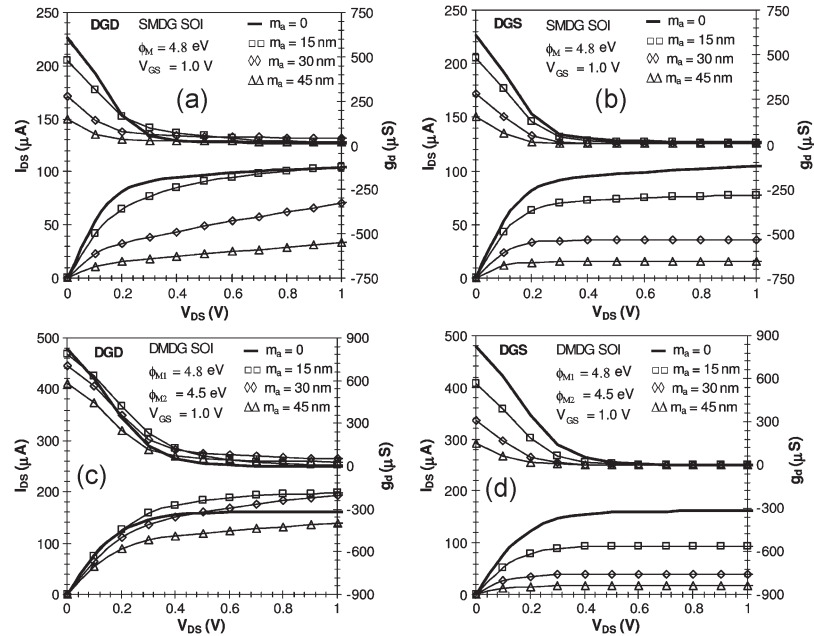


Fig. 8. I_{DS} - V_{DS} and g_d - V_{DS} curves for SMDG and DMDG SOI MOSFETs at $V_{GS} = 1.0$ V. (a) SMDG DGD configuration. (b) SMDG DGS configuration. (c) DMDG DGD configuration. (d) DMDG DGS configuration.

in the misalignment region, inversion charge emerges as the outcome of the fringing field and not by the normal gate field action. The effect of normal gate field in creating the inversion charge is more significant in comparison to what emerges as a result of the fringing field. This leads to a reduction in inversion charge density in the misaligned region that accounts for the reduced g_m and I_{DS} . For the DGS configuration of DMDG, I_{DS}

and g_m consistently decrease with the increase in misalignment similar to that of SMDG, whereas for the DGD configuration, it shows an inconsistent behavior. This is because, with the increase in misalignment, the effect of work-function variation decreases in the DGS configuration, whereas this effect increases in the DGD configuration that shows shielding of the misalignment effect in the DGD configuration.

Fig. 8 shows the simulated variations of drain current and drain conductance with drain-to-source bias for the SMDG and DMDG structures. As misalignment increases, drain current and drain conductance decrease for the DGD and DGS configurations in the SMDG structure. The DGS configuration of the DMDG structure shows larger effect of misalignment following the similar trend as in the SMDG structure, whereas the DGD configuration of the DMDG structure shows an inconsistent behavior in comparison to the SMDG structure.

V. CONCLUSION

The effect of gate misalignment in the DMDG SOI n-MOSFET has been examined for the first time by developing an analytical surface potential model and by using 3-D ATLAS device simulation. Although DMDG enhances the device performance, i.e., improves SCEs, increases drain current and transconductance, etc., a slight gate misalignment in DMDG causes degradation in device performance in comparison to SMDG devices. With the increase in misalignment for the DGS configuration of the DMDG device, threshold voltage consistently increases, similar to SMDG devices. However, for the DGD configuration, threshold voltage decreases with the increase in misalignment. This indicates that a misalignment toward the drain side (DGD configuration) is beneficial for switching applications because it leads to a reduction in the threshold voltage of the device, whereas a misalignment toward the source side (DGS configuration) leads to degraded device characteristics in terms of increased threshold voltage. Moreover, with the increase in misalignment for the DGS configuration, I_{DS} and g_m consistently decrease, similar to SMDG devices, whereas for the DGD structure, it shows an inconsistent behavior. Drain current and drain conductance also decrease with the increase in misalignment for both DGD and DGS configurations of the SMDG structure. The DGS configuration of DMDG shows larger effect of misalignment following the similar trend as in the SMDG structure, whereas the DGD configuration of DMDG shows conflicting behaviors.

REFERENCES

- [1] R. Y. Yan, A. Ourmazd, and K. F. Lee, "Scaling the Si MOSFET: From bulk to SOI to bulk," *IEEE Trans. Electron Devices*, vol. 39, no. 7, pp. 1704–1710, Jul. 1992.
- [2] K. Shzuki, T. Tanaka, Y. Tosaka, H. Horie, and Y. Arimoto, "Scaling theory for double-gate SOI MOSFETs," *IEEE Trans. Electron Devices*, vol. 40, no. 12, pp. 2326–2329, Dec. 1993.
- [3] C. H. Wann, K. Noda, T. Tanaka, M. Yoshida, and A. Hu, "Comparative study of advanced MOSFET concepts," *IEEE Trans. Electron Devices*, vol. 43, no. 10, pp. 1742–1753, Oct. 1996.
- [4] D. J. Frank, S. E. Laux, and M. V. Fischetti, "Monte Carlo simulation of a 30 nm dual-gate MOSFET: How short can Si go?" in *IEDM Tech. Dig.*, 1992, pp. 553–556.
- [5] T. Endoh, M. Mori, H. Sakuraba, and F. Masuoka, "Analysis of short channel effects and proposal of scaling theory in fully-depleted double-gate SOI MOSFET," *Trans. Inst. Electron., Inform. Commun. Eng.*, vol. j82-C-II, no. 2, pp. 72–73, 1999. IEICE C-II.
- [6] T. Ernst, S. Cristoloveanu, G. Ghibaud, T. Ouisse, S. Horiguchi, Y. Ono, Y. Takahashi, and K. Murase, "Ultimate thin double-gate SOI MOSFETs," *IEEE Trans. Electron Devices*, vol. 50, no. 3, pp. 830–838, Mar. 2003.
- [7] C. C. Tsai, Y. J. Lee, J. L. Wang, K. F. Wei, I.-C. Lee, C.-C. Chen, and H.-C. Cheng, "High-performance top and bottom double-gate low-temperature poly-silicon thin film transistors fabricated by excimer laser crystallization," *Solid State Electron.*, vol. 52, no. 3, pp. 365–371, Mar. 2008.
- [8] T. M. Chung, B. Olbrechts, U. Sodervall, S. Bengtsson, D. Flandre, and J.-P. Raskin, "Planar double-gate SOI MOS devices: Fabrication by wafer bonding pre-patterned cavities and electrical characterization," *Solid State Electron.*, vol. 51, no. 2, pp. 231–238, Feb. 2007.
- [9] X. Lin, C. Feng, S. Zhang, W.-H. Ho, and M. Chan, "Characterization of double gate MOSFETs fabricated by a simple method on a recrystallized silicon film," *Solid State Electron.*, vol. 48, no. 12, pp. 2315–2319, Dec. 2004.
- [10] N. D. Jankovic and G. A. Armstrong, "Performance comparison of double gate SOI MOSFETs on highly-doped and near-intrinsic layers," *Microelectron. J.*, vol. 35, no. 8, pp. 647–653, Aug. 2004.
- [11] Y. Liu, K. Ishii, T. Tsutsumi, and M. Masahara, "Systematic electrical characteristics of ideal rectangular cross section Si-Fin channel double-gate MOSFETs fabricated by a wet process," *IEEE Trans. Nanotechnol.*, vol. 2, no. 4, pp. 198–204, Dec. 2003.
- [12] H. Liu, Z. Xiong, and S. K. O. Johnny, "Implementation and characterization of double-gate MOSFET using lateral solid-phase epitaxy," *IEEE Trans. Electron Devices*, vol. 50, no. 6, pp. 1552–1555, Jun. 2003.
- [13] T. Schulz, W. Rosner, E. Landgraf, L. Risch, and U. Langmann, "Planar and vertical double gate concepts," *Solid State Electron.*, vol. 46, no. 7, pp. 985–989, Jul. 2002.
- [14] Q. Chen, B. Agrawal, and J. D. Meindl, "A comprehensive analytical sub-threshold swing model for double-gate MOSFETs," *IEEE Trans. Electron Devices*, vol. 49, no. 6, pp. 1086–1089, Jun. 2002.
- [15] A. Kumar, K. O. S. Johnny, C. T. Nguyen, and K. K. Ping, "Kink-free polycrystalline silicon double-gate elevated channel thin-film transistors," *IEEE Trans. Electron Devices*, vol. 45, no. 12, pp. 2514–2520, Dec. 1998.
- [16] T. Tanaka, K. Suzuki, H. Horie, and T. Sugii, "Ultrafast operation of V_{th} -Adjusted $p^+ - n^+$ double-gate SOI MOSFETs," *IEEE Electron Device Lett.*, vol. 15, no. 10, pp. 386–388, Oct. 1994.
- [17] W. Long, H. Ou, J.-M. Kuo, and K. K. Chin, "Dual-material gate (DMG) field effect transistor," *IEEE Trans. Electron Devices*, vol. 46, no. 5, pp. 865–870, May 1999.
- [18] X. Zhou and W. Long, "A novel hetero-material gate (HMG) MOSFET for deep-submicron ULSI technology," *IEEE Trans. Electron Devices*, vol. 45, no. 12, pp. 2546–2548, Dec. 1998.
- [19] A. Chaudhry and M. J. Kumar, "Controlling short-channel effects in deep submicron SOI MOSFETs for improved reliability: A review," *IEEE Trans. Device Mater. Rel.*, vol. 4, no. 1, pp. 99–109, Mar. 2004.
- [20] M. J. Kumar and A. Chaudhry, "Two-dimensional analytical modeling of fully depleted dual-material gate (DMG) SOI MOSFET and evidence for diminished short-channel effects," *IEEE Trans. Electron Devices*, vol. 15, no. 4, pp. 569–574, Apr. 2004.
- [21] G. V. Reddy and M. J. Kumar, "A new dual-material double-gate (DMDG) nanoscale SOI MOSFET—Two-dimensional analytical modeling and simulation," *IEEE Trans. Nanotechnol.*, vol. 4, no. 2, pp. 260–268, Mar. 2005.
- [22] M. J. Kumar and G. V. Reddy, "Diminished short channel effects in nanoscale double-gate silicon-on-insulator metal-oxide-semiconductor field-effect-transistors due to induced back-gate step potential," *Jpn. J. Appl. Phys.*, vol. 44, no. 9A, pp. 6508–6509, Sep. 2005.
- [23] T. K. Chiang and M. L. Chen, "A new two-dimensional analytical model for short-channel symmetrical dual-material double-gate metal-oxide-semiconductor field effect transistors," *Jpn. Soc. Appl. Phys.*, vol. 46, no. 6A, pp. 3283–3290, Jun. 2007.
- [24] C. Yin and P. C. H. Chan, "Investigation of the source/drain asymmetric effects due to gate misalignment in planar double-gate MOSFETs," *IEEE Trans. Electron Devices*, vol. 52, no. 1, pp. 85–90, Jan. 2005.
- [25] J. Widiez, J. Lolivier, M. Vinet, T. Poiroux, F. Dauge, M. Mouis, and S. Deleonibus, "Experimental evaluation of gate architecture influence on DG SOI MOSFETs performance," *IEEE Trans. Electron Devices*, vol. 52, no. 8, pp. 1772–1779, Aug. 2005.
- [26] E. C. Sun and J. B. Kuo, "A compact threshold voltage model for gate misalignment effect of DG FD SOI NMOS devices considering fringing electric field effects," *IEEE Trans. Electron Devices*, vol. 51, no. 4, pp. 587–596, Apr. 2004.
- [27] C. Yin, P. C. H. Chan, and V. W. C. Chan, "Fabrication of raised S/D gate-all-around transistor and gate misalignment analysis," *IEEE Electron Device Lett.*, vol. 24, no. 10, pp. 658–660, Oct. 2003.
- [28] R. K. Sharma, R. Gupta, M. Gupta, and R. S. Gupta, "Graded channel architecture: The solution for misaligned DG FD SOI n-MOSFETs," *Semicond. Sci. Technol.*, vol. 23, no. 7, p. 075 041, Jul. 2008.
- [29] *ATLAS Users Manual*, Silvaco Int., Santa Clara, CA, 2003.
- [30] D. Querlioz, J. S. Martin, K. Huet, A. Bournel, V. Aubry-Fortuna, C. Chassat, S. Galdin-Retailleau, and P. Dollfus, "On the ability of the

particle Monte Carlo technique to include quantum effects in nano-MOSFET simulation," *IEEE Trans. Electron Devices*, vol. 54, no. 9, pp. 2232–2242, Sep. 2007.

- [31] S. S. Chen and J. B. Kuo, "Deep submicrometer DG FD SOI PMOS devices: A concise short-channel effect threshold voltage model using a quasi-2D approach," *IEEE Trans. Electron Devices*, vol. 43, no. 9, pp. 1387–1393, Sep. 1996.
- [32] H. Y. Wong, K. Shin, and M. Chan, "The gate misalignment effects of the sub-threshold characteristics of sub-100nm DG-MOSFETs," in *IEDM Tech. Dig.*, 2002, pp. 91–94.



Rupendra Kumar Sharma was born on July 1, 1984. He received the B.Sc. and M.Sc. degrees in physics from Chaudhary Charan Singh (CCS) University, Meerut, India, in 2003 and 2005, respectively. He is currently working toward the Ph.D. degree in electronics in the Semiconductor Devices Research Laboratory, Department of Electronic Science, University of Delhi South Campus, New Delhi, India.

He has authored or coauthored nine publications in international and national journals and conference proceedings. His research interests include modeling, simulation, and characterization of gate misalignment effect in double-gate and dual-material double-gate SOI MOSFETs.



Ritesh Gupta was born in Delhi, India, on July 20, 1976. He received the B.Sc. and M.Sc. degrees in physics and the Ph.D. degree in microelectronics from the University of Delhi, Delhi, in 1997, 1999, and 2003, respectively.

He joined the Semiconductor Devices Research Laboratory, Department of Electronic Science, University of Delhi South Campus, New Delhi, India, in 1999 as a Junior Research Fellow, where he is currently a Research Associate on a Department of Science and Technology (DST)-sponsored project.

His research interests include modeling and simulation of Si/SiC/InP MESFET/MOSFET/HEMT devices for high-frequency applications. He has published 28 technical papers in various journals and conferences.



Mridula Gupta (M'01) received the B.Sc. degree in physics, the M.Sc. degree in electronics, the M.Tech. degree in microwave electronics, and the Ph.D. degree in optoelectronics from the University of Delhi, Delhi, India, in 1984, 1986, 1988, and 1998, respectively.

Since 1989, she has been with the Department of Electronic Science, University of Delhi South Campus, New Delhi, India, where she was previously a Lecturer and is currently an Associate Professor and with the Semiconductor Devices Research Laboratory. She has authored or coauthored approximately 214 publications in international and national journals and conference proceedings. She has supervised 14 Ph.D. students. She contributed the chapter entitled "MOSFET Modeling" in the *Encyclopedia on RF and Microwave Engineering* (Wiley, 2005). Her current research interests include modeling and simulation of MOSFETs, MESFETs, and HEMTs for microwave-frequency applications.

Dr. Gupta is a Fellow of the Institution of Electronics and Telecommunication Engineers (India) and a Life Member of the Semiconductor Society of India. She was the Secretary of the 2004 Asia-Pacific Microwave Conference, New Delhi, held in December 2004.



R. S. Gupta (SM'81) received the B.Sc. and M.Sc. degrees from Agra University, Agra, India, in 1963 and 1966, respectively, and the Ph.D. degree in electronic engineering from the Institute of Technology, Banaras Hindu University, Varanasi, India, in 1970.

In 1971, he was with Ramjas College, University of Delhi, Delhi, India. In 1987, he was with the Department of Electronic Science, University of Delhi South Campus, New Delhi, India, as a Reader and later as a Professor from 1997 to 2008, where he is currently an Emeritus Scientist and with the

Semiconductor Devices Research Laboratory. He heads several major research projects sponsored by the Ministry of Defence, the Department of Science and Technology, the Council of Science, and the Industrial Research and University Grants Commission. In 1988, he was a Visitor with the University of Sheffield, Sheffield, U.K., under the ALIS Link exchange program and also visited several U.S. and Spanish universities in 1995 and 1999, respectively. He also visited the Czech Republic in August 2003; Korea in November 2003; Rensselaer Polytechnic Institute, Troy, NY, in August 2004; and China in December 2005. He has authored or coauthored over 472 papers in various international and national journals and conference proceedings. He contributed the chapter entitled "MOSFET Modeling" in the *Encyclopedia on RF and Microwave Engineering* (Wiley, 2005). He has supervised 47 Ph.D. students. His current interests and activities include modeling of SOI submicrometer MOSFETs and LDD MOSFETs, modeling and design of HEMTs, hot-carrier effects in MOSFETs, and modeling of GaAs MESFETs for high-performance microwave and millimeter-wave circuits and quantum-effect devices.

Dr. Gupta was an Executive Member of the IEEE Electron Devices Society/Microwave Theory and Techniques Society Chapter of the IEEE India Council. His name also appeared in the Golden List of the IEEE TRANSACTIONS ON ELECTRON DEVICES in December 1998, 2002, and 2004. He is a Fellow of the Institution of Electronics and Telecommunication Engineers (India), a Life Member of the Indian Chapter of the International Centre for Theoretical Physics, and a Life Member of the Semiconductor Society of India. He was the Secretary of ISRAMT'93 and the 1996 Asia-Pacific Microwave Conference (APMC'96), and the Chairman of the Technical Programme Committee of APMC'96. He edited the proceedings of both of these international conferences. He was the Chairman of APMC'2004 held in New Delhi in December 2004. He has been listed in *Who's Who in the World*.